American Computer Science League

2020-2021 ● Contest 4: Solutions ● Intermediate Division

1. Graph Theory

Cubing the adjacency matrix will give the number of paths of length 3 in the directed graph.

$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}^3 \end{array} = \begin{bmatrix} 3 & 0 & 3 & 3 \\ 3 & 1 & 3 & 1 \\ 3 & 0 & 2 & 3 \\ 3 & 0 & 3 & 2 \end{bmatrix}$$

A, C and D do not have paths of length 3 to vertex B.

1. A, C, D (D)

2. Graph Theory

There are 7 cycles from B in the directed graph: BAB, BCAB, BDB, BADB, BAEDB, BCADB, and BCAEDB.

2. 7 (C)

3. **Digital Electronics**

The Boolean expression represented by the digital circuit is:

$$\begin{aligned} & \overline{(A + (\overline{AB})\,(B + C))}\,C \\ &= (\overline{A}\,\overline{((\overline{AB})\,(B + C))})\,C \\ &= \overline{A}\,C\,((\overline{\overline{AB}}) + (\overline{B + C})) \\ &= \overline{A}\,C\,(AB + \overline{B}\,\overline{C}) \\ &= A\,\overline{A}\,B\,C + \overline{A}\,\overline{B}\,C\,\overline{C} \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

Since the expression is always FALSE, none, or 0, make it TRUE.

3. 0 (A)

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INTERMEDIATE DIVISION SOLUTIONS

4. Digital Electronics

The Boolean expression represented by the digital circuit is:

$$(A \oplus B)(B \oplus C)$$
$$= (\overline{A}B + A\overline{B})(\overline{B}C + B\overline{C})$$
$$= \overline{A}\,B\,\overline{B}\,C + \overline{A}\,B\,B\,\overline{C} + A\,\overline{B}\,\overline{B}\,C + A\,\overline{B}\,B\,\overline{C}$$
$$= 0 + \overline{A}\,B\,\overline{C} + A\,\overline{B}\,C + 0$$
$$= \overline{A}\,B\,\overline{C} + A\,\overline{B}\,C$$

To be TRUE one of the terms must be TRUE.

$\overline{A}\,B\,\overline{C} = 1 \Rightarrow A = 0,\ B = 1,\ C = 0 \Rightarrow (0, 1, 0)$

$A\,\overline{B}\,C = 1 \Rightarrow A = 1,\ B = 0,\ C = 1 \Rightarrow (1, 0, 1)$

4. (0, 1, 0) (1, 0, 1) (E)

5. Assembly Language

This is a translation of the assembly language program:

```
x = 21
y = 49
while y != 0
        r = int(x / y) * y
        r = x - r
        x = y
        y = r
end while
output x
```

This is the Euclidean Algorithm to find the greatest common divisor of two numbers. The GCF of 21 and 49 is 7.

5. 7 (B)